

BALAJI RADHAKRISHNAN PADMANABHAN

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SUMMARY

Software engineer with three years of experience delivering production-ready healthcare software solutions using Python, databases, and automation. Pursuing an M.S. in Software Engineering at Arizona State University, graduating in May 2027, and seeking full-time software engineering roles starting in 2027.

EDUCATION

Arizona State University, Ira A. Fulton Schools of Engineering

Tempe, AZ

Master of Science in Software Engineering | GPA: 3.56/4.00

*Aug 2025 – **Expected May 2027***

Relevant Coursework: Advanced Data Structures and Algorithms, Data Processing at Scale, Web Development

SRM Institute of Science and Technology

Chennai, India

Bachelor of Technology in Computer Science and Engineering | CGPA: 8.64/10.00

Jun 2018 – May 2022

TECHNICAL SKILLS

Languages: Python, SQL, C#, TypeScript, JavaScript, Bash, PowerShell

Backend & APIs: FastAPI, Flask, REST APIs, SQLAlchemy, Alembic, Pydantic, ASP.NET Core, Entity Framework Core

Frontend: React, Angular

Databases & Data: PostgreSQL, MySQL, Pandas, NumPy, ELT/ETL pipelines, Tableau

Cloud & Infrastructure: Google Cloud (Cloud Run, Cloud SQL, Cloud Build, Cloud Scheduler, Secret Manager), AWS (S3, RDS), Docker, Kubernetes, Terraform, Linux

CI/CD & Testing: Git, GitHub Actions, Jenkins, Pytest, xUnit, Testcontainers, Selenium, Playwright, Postman

Applied AI: Google ADK, Vertex AI, Gemini, OpenAI API

PROFESSIONAL EXPERIENCE

Tata Consultancy Services | Client: Cigna Healthcare

Chennai, India

Systems Engineer (promoted from Assistant Systems Engineer)

Sep 2022 – Jul 2025

- Built and extended Python tooling for a healthcare RFP platform used by enterprise clients, wiring execution and reporting into Jenkins CI/CD pipelines across 50+ Agile sprints in a 12-member team.
- Wrote multi-table SQL and Pandas checks over PostgreSQL and AWS S3 data pipelines verifying transformations end to end for nulls, duplicates, schema drift, and source-to-target mismatches before data reached downstream consumers.
- Worked across a distributed system spanning Salesforce, Pega, a Next.js portal, REST APIs, and downstream ETL jobs, tracing payloads, logs, and transformations between services to resolve issues at system boundaries.
- Exercised REST API contracts in Postman across service versions, covering authentication, payload schemas, and error handling, and confirmed reporting matched the underlying database through SQL.

Arizona State University – Enterprise Technology, MIX Center

Mesa, AZ

IT Deskside Support Assistant (Part-time)

Mar 2026 – Present

- Automate software installation, configuration, and deployment validation with PowerShell, Bash, and Linux/macOS tooling across managed Windows and macOS fleets.
- Image and deploy endpoints through SCCM, Intune, and Jamf, administer Active Directory access across the user lifecycle, and troubleshoot endpoint and TCP/IP connectivity.

PROJECTS

TriageZero: Agentic Release Engineering Platform

2026

Google All Things Agentic Hackathon | Python, FastAPI, Google ADK, Vertex AI, Cloud Run, React

- Built a FastAPI backend and React dashboard that receive regression-failure evidence through a versioned REST contract and return a diagnosis with severity, release risk, and a recommended action, correct on all 84 evaluation cases within the supported defect class.
- Orchestrated a seven-stage Google ADK workflow calling Gemini through Vertex AI with read-only evidence tools, validating each response against a closed schema, falling back to a deterministic analyzer, and keeping release-risk policy outside the model.
- Deployed four Cloud Run services and two scheduled jobs backed by Cloud SQL PostgreSQL, with builds automated through Cloud Build.
- Controlled ingestion with token authentication, request validation, and idempotency-key deduplication, and recorded model, prompt, and schema versions with latency, token usage, and fallback reason so any run can be investigated afterwards.

Inventory and Equipment Lending API

2026

FastAPI, PostgreSQL, SQLAlchemy, Alembic, Docker, Pytest

- Built a lending service tracking quantity-based consumables and individually serialized equipment through checkout, return, and stock adjustment for manager, supervisor, and staff roles.
- Designed a normalized eight-table schema that records stock changes in an append-only movement ledger, so current quantities are derived from recorded movements rather than mutable fields.
- Enforced loan and inventory consistency in the database through check constraints, row-level locking on units during checkout, and a partial unique index that permits one active loan per unit.
- Secured endpoints with JWT authentication and role-based access, versioned the schema through Alembic migrations, and packaged the service with Docker alongside Pytest tests.

PocketLense: Personal Finance Application

2026

C#, ASP.NET Core, Angular, PostgreSQL, Entity Framework Core, Docker, Nginx, OpenAI API

- Built an Angular application backed by an ASP.NET Core API that uses Entity Framework Core over PostgreSQL to manage transactions, monthly budgets, and recurring subscription tracking.
- Imported statements from CSV and Excel through user-defined column mapping, parsed text-based PDF statements through a separate path, and applied duplicate detection with all-or-nothing processing so a failed import leaves no partial data.
- Protected user data with JWT authentication and queries scoped to the authenticated user.
- Implemented an OpenAI-powered budgeting assistant with read-only reporting tools scoped to the authenticated user, so it explains spending and budget position without modifying financial records.
- Packaged the frontend and backend as separate containers with Nginx serving the Angular build and forwarding API requests, and ran xUnit tests, PostgreSQL integration tests through Testcontainers, and build checks in GitHub Actions.

Global Superstore Data Pipeline

2026

Python, PostgreSQL, SQL, Docker, Tableau, Gemini API

- Loaded 51,290 retail records into PostgreSQL and carried out the transformations in the database rather than in application code.
- Made loads transactional and idempotent to avoid duplicate rows when the pipeline is rerun, with Docker Compose providing a reproducible execution workflow.
- Documented six reproducible source-data quality issues in the raw extract and implemented the corresponding cleaning and transformation logic, including recovery of malformed order dates.
- Exposed SQL views used by both Tableau dashboards and a Gemini question-answering layer that attributes its answers to the underlying sources.

PUBLICATION

Designing a Dynamic Framework for People Counting Using YOLO-PC | AIP Conf. Proceedings, 2024 | ICCCN 2022